Digital-camera exposure for film scanning

Why use Camera-41_exp?

Christer Törnkvist

Camera-41_exp helps set the digital camera exposure when digitising C-41 colour and black-and-white (B&W) film negatives with a VALOI setup. At a fixed aperture, it compares RAW captures of clear film base plus fog to find a shutter speed that gives a strong signal with room before clipping. Too much exposure can clip a sensor channel. Too little leaves a weak signal, making noise more visible when the scan is brightened.

Avoid clipping in the RAW capture

An empty frame containing film base plus fog (FB+F) normally transmits more light than denser image areas of a negative. It is therefore an important exposure reference. If the shutter time is too long, one colour channel may reach its limit. Measurable information is then lost; making the preview darker cannot restore it.
Exposure reads the original Bayer values and compares signal and headroom relative to LibRaw's linearity limit in R, G1, G2 and B. The aim is a strong usable signal with margin, not the brightest image on screen.

What the app does

It compares RAW files at different shutter times, normally chooses an area away from the edges and plots signal and clipping margin. It proposes the longest actually measured time that meets its criteria. A proposal may need further test captures.

What it does not do

Camera-41_exp does not measure dark noise or full S/N and does not create a production qualification. It does not control the camera. The exposure choice concerns the digital capture, not the original analog exposure when the film was photographed.

Quick start

1. Open Camera-41_exp from Applications.
2. Use Add FB+F RAWs… to load an exposure bracket of the same empty film sample.
3. Enter Light setup and Film base + fog. Tick Same FB+F sample in all RAWs.
4. Press Analyse RAW images and read Results and Exposure comparison.
5. Verify the selected time with a fresh capture.

christer@borgasundsfotografiska.se
https://github.com/chto0703009/Camera-41_exp

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Installation and startup

The standalone macOS app includes Python and LibRaw. The interface is in English.

1. Open the disk image

Open Camera-41_exp-1.0.1-python-macos.dmg. Drag the Apple Silicon app to Applications on an M-series Mac, or choose the Intel app for an Intel Mac.

2. Open the app

Eject the disk image and open the app from Applications. The apps are ad hoc signed. The disk image has no Apple notarisation.

Verification

The full four-file Gold 200 workflow was exercised on Apple Silicon. Earlier Intel builds were exercised through Rosetta. Physical Intel testing remains outstanding.

RAW files and diagnostics

Add your own film-base RAWs with Add FB+F RAWs…. Open log shows the local diagnostic log. Quit the app before replacing it.

Check the download

Compare the installer checksum with the accompanying SHA-256 file. On macOS use shasum -a 256 followed by the installer filename. See INSTALL.md for source setup.

Native RAW reader

The standalone app carries the LibRaw reader for its CPU architecture. It reads undemosaiced RAW sensor values. The exposure assessment expects an RGGB mosaic.

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Capture files and select an area

Use an empty film frame containing base plus fog from the film being scanned. A negative with a scene, text, perforations or a holder edge is not a suitable empty FB+F area.

Keep the setup fixed

Use the same camera, ISO, aperture, light mode and geometry throughout the bracket. Choose the aperture first and change only shutter time. Start a separate comparison if the light, film sample or setup changes. Light setup and Film base + fog document what was measured.

Add and preview RAW files

Add FB+F RAWs… accepts one or more files. Later selections are added to the list; the same path is not duplicated. The file chooser remembers the latest RAW folder across app restarts. Remove selected removes a row from the bracket, not the original file.
Select a row and press Preview selected to inspect it. RAW preview is a display image with brightness adjustment; screen brightness does not determine whether the measurement passes.

Automatic area

Automatic area is enabled by default. The central 70 percent of visible width and height is used, leaving a 15 percent border on each side. The same area is measured throughout the bracket. The app does not hunt for a small favourable patch to make an excessive exposure pass.
A per-Bayer-channel screen checks for substantial structure, large illumination differences or insufficient signal. This assists inspection but cannot reliably identify the physical contents of a film frame.

When a warning requires adjustment

When a warning appears, the app shows the affected image and outlines the proposed area in yellow. Drag a new rectangle over clean empty film or edit X, Y, Width and Height in native visible sensor pixels. Use the same area across the bracket.

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Interpret the results

Successful RAW reading and a passing exposure are different outcomes. FAIL can be a valid measurement from a file that was read successfully.

PASS, REVIEW and FAIL

PASS: every channel retains at least 1/3 EV headroom at the 99.9th percentile, with too few values at/above the linearity limit to trigger FAIL.
REVIEW: no channel triggers FAIL, but at least one retains less than 1/3 EV margin. Do not increase exposure; try a slightly shorter time.
FAIL: at least one channel has 0.01 percent or more of its measured values at/above the limit. First check the sample and measurement area. If they are correct, try a shorter shutter time.
The reference is the RAW linearity limit after black subtraction, not a JPEG histogram. Loss of linearity can matter before hard sensor saturation.
LOW marks a channel whose median lies more than 8 EV below the clipping reference. Some cameras use LibRaw’s white level as the reference.

Plots and table

Exposure comparison uses 1/3 EV shutter steps and camera-style times such as 1/20, 1/15 and 1/10. The upper chart shows median RAW signal above black in sensor codes. The lower chart shows headroom before the RAW linearity limit in EV. At 0 EV the measured percentile reaches the limit. The dashed 1/3 EV line marks the minimum headroom for PASS. Results lists the file, time, status, signal, headroom and fraction at or above the limit for each channel.

A candidate may need more evidence

The proportionality check compares signal per second to its median for each Bayer channel, using at least three distinct passing times spanning one stop or more. A deviation above 5% in any channel triggers a review before using the candidate. In the Gold 200 test, 1/20, 1/15 and 1/10 s passed the clipping criteria. At 1/8 s the green channels triggered REVIEW. The candidate was 1/10 s with a proportionality warning (maximum deviation 5.23%).

Choose a time with margin

Choose a measured time lists the times for which all replicates passed the clipping criteria. Review any proportionality warning before using the candidate. Verify your chosen time with a fresh FB+F capture. If the results contradict each other, the app asks you to repeat the series before choosing a time.

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After the test, logging and help

Camera-41_exp works independently. No JSON file is transferred from Camera-41_exp to PRE or NEG, and neither program needs to be open to use it.

Continue in PRE with current captures

Set the digital camera to the chosen time yourself. Capture the current film's empty FB+F frame and two separate dark RAWs with the same camera, ISO, shutter time and aperture. For the dark frames, block light from reaching the sensor and use relevant matching capture conditions.
Load those files directly in PRE tab 7. FB+F and the dark pair can be inspected with a common sensor area and used for noise and S/N measurement. PRE includes its measured result in the film characterisation delivered to NEG. Camera-41_exp does not export that measurement and need not be installed for PRE to work.

If something looks wrong

Select a file and press Preview selected. For a failed analysis, read the status message and inspect the local log. Check that the RAW file and its drive are still available.

Log and retained information

Open log opens the local diagnostic log. The standalone app writes ~/Library/Logs/Camera-41_exp/camera41_exp.jsonl. Source runs write logs/camera41_exp.jsonl. The app remembers the last RAW folder. It opens with an empty bracket after restart.

Requirements, edition and licences

This handbook covers the Python app in v1.0.1. Camera-41_exp uses GPL-3.0-or-later. See LICENSE, LICENSE_NOTICE.md and the bundled third-party notices.

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